25. Justify the rule of universal modus tollens by showing that the premises $\forall x(P(x) \rightarrow Q(x))$ and $\neg Q(a)$ for a particular element a in the domain, imply $\neg P(a)$.

Soln: Premise: 1. $\forall x(P(x) \rightarrow Q(x))$ 2. $\neg Q(a)$ for some a in the domain

Conclusion: $\neg P(a)$

Steps	Reason
1. $\forall x(P(x) \rightarrow Q(x))$	Premise
2. $\neg Q(a)$ for some a in the domain	Premise
3. $P(a) \rightarrow Q(a)$	Universal instantiation on 1
4. $\neg P(a)$	Modus tollens on 2,3

26. Justify the rule of universal transitivity, which states that if $\forall x(P(x) \rightarrow Q(x))$ and $\forall x(Q(x) \rightarrow R(x))$ are true, then $\forall x(P(x) \rightarrow R(x))$ is true, where the domains of all quantifiers are the same.

Soln: Premise: 1. $\forall x(P(x) \rightarrow Q(x))$ 2. $\forall x(Q(x) \rightarrow R(x))$

Conclusion: $\forall x(P(x) \rightarrow R(x))$

Steps	Reason
1. $\forall x(P(x) \rightarrow Q(x))$	Premise
2. $\forall x(Q(x) \rightarrow R(x))$	Premise
3. $P(a) \rightarrow Q(a)$	Universal instantiation on 1
4. $Q(a) \rightarrow R(a)$	Universal instantiation on 2
5. $P(a) \rightarrow R(a)$	Hypothetical syllogism on 3,4
6. $\forall x(P(x) \rightarrow R(x))$	Universal generalization on 5